Jaihoon Kim

Website | LinkedIn | Email: jh27kim@kaist.ac.kr

RESEARCH INTERESTS

Generative AI, Diffusion Model, Flow Model, Discrete Diffusion

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) — Sep 2024 – Feb 2028 (Expected)
Ph.D. in Computer Science — Advisor: Minhyuk Sung
Daejeon, South Korea

Korea Advanced Institute of Science and Technology (KAIST) — Feb 2023 – Aug 2024
M.S. in Computer Science — Advisor: Minhyuk Sung
Daejeon, South Korea

Hanyang University — March 2015 – Aug 2021
B.S. in Electrical Engineering
Seoul, South Korea

WORK EXPERIENCE

Netflix (Los Gatos, United States) — June 2026 – Dec 2026
Research Scientist (Internship)

Amazon Web Services (Santa Clara, United States) — July 2025 – Nov 2025
Research Scientist (Internship)

Samsung Electronics (Hwasung, South Korea) — July 2021 – Jan 2023
Cloud Engineer (Full Time)

Samsung Electronics (Hwasung, South Korea) — July 2020 – July 2020
Cloud Engineer (Internship)

PUBLICATIONS

1. Contrastive Distribution Matching for Amortized Sequential Monte Carlo in Discrete Diffusion
Jaihoon Kim, Taehoon Yoon, Prin Phunyaphibarn, Seungjun Kim, Morteza Mardani, Minhyuk Sung
NeurIPS 2026

2. NoiseTilt: Noise-Tilted Reverse Kernels for Diffusion Reward Alignment
Jisung Hwang, Yunhong Min, **Jaihoon Kim**, I-Chao Shen, Minhyuk Sung
ECCV 2026

3. MatLat: Material Latent Space for PBR Texture Generation
Kyeongmin Yeo, Yunhong Min, **Jaihoon Kim**, Minhyuk Sung
CVPR 2026 (Highlight)

4. Demystifying Transition Matching: When and Why It Can Beat Flow Matching
Jaihoon Kim, Rajarshi Saha, Minhyuk Sung, Youngsuk Park
AISTATS 2026

5. Inference-time scaling for flow models via stochastic generation and rollover budget forcing
Jaihoon Kim*, Taehoon Yoon*, Jisung Hwang*, Minhyuk Sung (* equal contribution)
NeurIPS 2025

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6. Moment-and Power-Spectrum-Based Gaussianity Regularization for Text-to-Image Models
   Jisung Hwang, **Jaihoon Kim**, Minhyuk Sung
   NeurIPS 2025

7. Moment-and Power-Spectrum-Based Gaussianity Regularization for Text-to-Image Models
   Prin Phunyaphibarn, Phillip Y Lee, **Jaihoon Kim**, Minhyuk Sung
   WACV 2026

8. StochSync: Stochastic Diffusion Synchronization for Image Generation in Arbitrary Spaces
   Kyeongmin Yeo*, **Jaihoon Kim***, Minhyuk Sung (* equal contribution)
   ICLR 2025

9. SyncTweedies: A General Generative Framework Based on Synchronized Diffusions
   Jaihoon Kim*, Juil Koo*, Kyeongmin Yeo*, Minhyuk Sung (* equal contribution)
   NeurIPS 2024

10. Paired-Point Lifting for Enhanced Privacy-Preserving Visual Localization
    Chunghwan Lee, **Jaihoon Kim**, Chanhyuk Yun, Je Hyeong Hong
    CVPR 2023

PREPRINTS

1. sDMD: Sinkhorn Distribution Matching Distillation
   Jaihoon Kim, Lingxiao Li, Soshi Shimada, Roshni Cooper, Minhyuk Sung, Yiwei Zhao
   2026

2. Unbalanced W-Flow: Unbalanced Wasserstein Drifting under Finite-Batch Marginal Fluctuations
   Jongwook Jeon, Mingue Park, Jiwon Choi, **Jaihoon Kim**, Yunhong Min, Minhyuk Sung
   2026

TEACHING EXPERIENCE

CS479: Machine Learning for 3D Data — School of Computing, KAIST
Course Assistant — Spring 2025

CS492(D): Diffusion Models and Their Applications — School of Computing, KAIST
Course Assistant — Fall 2024

CS580: Computer Graphics — School of Computing, KAIST
Course Assistant — Spring 2024

CS479: Machine Learning for 3D Data — School of Computing, KAIST
Course Assistant — Fall 2023

SCHOLARSHIPS AND ACHIEVEMENTS

Master's Degree Outstanding Paper — Summer 2024
KAIST

Grand Prix (2nd place) x, X-Corps Festival — Spring 2020
Ministry of Science and ICT

Jaihoon Kim

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Honors — Fall 2019
University of Texas at Austin

Dean's List — Spring 2019
Hanyang University

ACADEMIC SERVICE

Workshop Organizer

BeNTo: Beyond Next-Token Prediction (NeurIPS 2026)

Reviewer

ICLR 2024-2027
NeurIPS 2024- 2026
ICCV 2025, ECCV 2026
CVPR 2026

SKILLS

Programming Languages	Python, C++
Frameworks	PyTorch
Languages	Korean (Native), English (Fluent)